

Introduction to Fractions

1 Background Knowledge Check

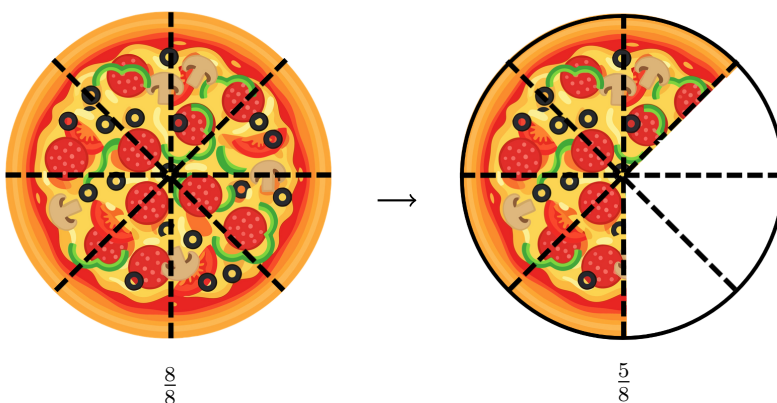
1. Understanding of addition, subtraction, multiplication, and division
2. Familiarity with multiplication tables.

2 What is a Fraction?

A fraction represents a part of something. It is written as two numbers stacked on top of each other, separated by a line. For example, $\frac{1}{2}$. This fraction represents a half of a whole. The number on top (in this case, 1) is called the **numerator**, and the number on the bottom (in this case, 2) is called the **denominator**. The denominator tells us how many equal parts the whole is divided into, while the numerator tells us how many of those parts we have.

$$\frac{\text{numerator}}{\text{denominator}}$$

Let's say you have a pizza that you cut into 8 equal slices, and out of these slices, you eat 3. The fraction that represents the amount of pizza you ate is $\frac{3}{8}$, where 3 is the numerator (the number of slices you ate), and 8 is the denominator (the total number of slices). The fraction that represents the amount of remaining slices is $\frac{5}{8}$.



3 Fractions and Division

Fractions can also be thought of as division. For example, $\frac{1}{2}$ can also be written as $1 \div 2$. The fraction $\frac{1}{2}$ represents a half of a whole, whereas $1 \div 2$ represents the result of dividing 1 into 2 equal parts. Both of them provide us with a value of a half, and are equivalent expressions.

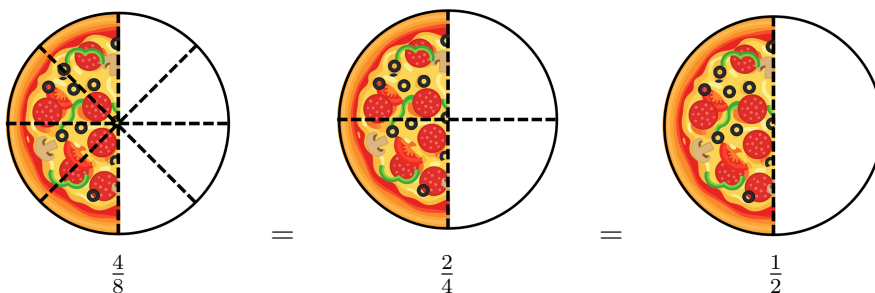
4 Equivalent Fractions and Simplifying Fractions

Two different fractions may represent the same value. For example, $\frac{1}{2}$ and $\frac{2}{4}$ are equivalent fractions because they both represent the same value of a half. Usually, we like to keep our fractions in what we call "their simplest form". To simplify a fraction, we need to find the greatest common factor between the numerator and denominator, and divide both of them by that factor. For example, to simplify $\frac{2}{4}$, we find the greatest common factor of 2 and 4, which is 2. We then divide both the numerator and denominator by 2, giving us $\frac{1}{2}$, which is the simplest form of the fraction:

$$\frac{2}{4} = \frac{2 \div 2}{4 \div 2} = \frac{1}{2}$$

Usually, we write this as if we cancelled both the numerator and the denominator and replaced it with the result:

$$\frac{2^1}{4^2} = \frac{1}{2}$$



5 Types of Fractions

1. **Proper Fractions:** A fraction whose numerator is smaller than its denominator. For example, $\frac{2}{5}$ is a proper fraction because $2 < 5$.
2. **Improper Fractions:** A fraction whose numerator is bigger than its denominator. For example, $\frac{5}{3}$ is an improper fraction because $5 > 3$.
3. **Mixed Fractions:** An improper fraction can be converted into a mixed fraction, which is a combination of a whole number and a proper fraction. For example, $\frac{5}{2}$ has 2 whole parts ($4 \div 2 = 2$) and a remaining part of $\frac{1}{2}$, so it can be written as $2\frac{1}{2}$.

6 Converting Between Different Types of Fractions

1. **Improper to mixed:**

To convert improper fractions to mixed fractions, we first separate the numerator into two numbers. The first number is the biggest number that is equal to or lower than the numerator AND divisible by the denominator. The second number is the difference between the numerator and the number we just found. For example:

$$\frac{5}{2}$$

$\frac{5}{2}$ has a numerator of 5, therefore the biggest number that is equal to or lower than 5 and also divisible by 2 is 4. The difference between 5 and 4 is 1. Therefore, we split our numerator as follows:

$$\frac{5}{2} = \frac{4 + 1}{2}$$

Next, we split our fraction in two using addition:

$$= \frac{4}{2} + \frac{1}{2}$$

Then, we simplify the improper fraction into a whole number:

$$= 2 + \frac{1}{2}$$

Finally, we write this as a mixed fraction by taking the whole part and writing it in front of our fraction:

$$= 2\frac{1}{2}$$

Converting from Improper to Mixed

- (a) Start by splitting the numerator into a sum of two numbers:
 - i. The biggest number that is both 1) equal to or lower than the current numerator AND 2) divisible by the denominator.
 - ii. The difference between the first number and the current numerator
- (b) Split the fraction into a sum of two fractions (One will be improper and the other proper)
- (c) Convert the improper fraction into a whole by dividing
- (d) Write the whole in front of the proper fraction

AND DONE!

2. Mixed to improper:

Let's start with the following example:

$$3\frac{4}{5}$$

To turn this into an improper fraction, we are going to keep our denominator the same and our numerator becomes a sum of our whole times the denominator and the current numerator:

$$\frac{3 \times 5 + 4}{5}$$

Then we just solve and simplify:

$$\frac{15 + 4}{5} = \frac{19}{5}$$

and done!

This can also be done by reversing the steps we took to turn the improper fraction

into a mixed fraction, but that requires knowing how to add two fractions, see the "Operations with Fractions" SLS document to learn more!

Converting from Mixed to Improper

Starting with a mixed fraction with any numbers in place of a, b and c:

$$a\frac{b}{c}$$

Our improper fraction can be found by solving:

$$\frac{a \times c + b}{c}$$